

## PART 1 - GENERAL

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**21 10 00 Part 1.1.1.A** This section specifies the requirements for fire detection & vesda for the Meridian Data Centre Campus - Phase 1.

### 1.1 Summary

**21 10 00 Part 1.1.1.B** The scope of supply includes all equipment, accessories, and installation materials required for a complete and operational system.

**21 10 00 Part 1.2.1.A** The following standards shall apply: NFPA 72, EN 54-20 and all applicable local codes.

### 1.2 References

**21 10 00 Part 1.3.1** The Contractor shall submit the equipment submittal package referencing the current active project specification revision Rev 2024. Submittals referencing an incorrect or outdated specification revision shall be returned without review.

### 1.3 Submittal Requirements

**21 10 00 Part 1.3.2.A** Submittals shall include certified manufacturer product data sheets, performance test reports, dimensional drawings, and a clause-by-clause compliance matrix.

**21 10 00 Part 1.4.1.A** Manufacturer shall have a minimum of 10 years experience in the design and manufacture of fire detection & vesda, with at least 5 reference installations in Tier III or above data centres.

### 1.4 Quality Assurance

**21 10 00 Part 1.4.1.B** Manufacturing facility shall hold current ISO 9001 and ISO 14001 certifications.

**21 10 00 Part 1.5.1.A** The manufacturer shall provide a minimum of 24 months on-site warranty from the date of commissioning.

### 1.5 Warranty and Spares

**21 10 00 Part 1.5.1.B** A recommended spare parts list with pricing shall be submitted for Consultant review.

## PART 2 - PRODUCTS

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### 2.1 Acceptable Manufacturers and System Configuration

**21 10 00 Part 2.1.1.A** Acceptable manufacturers: FireLite or approved equivalent meeting all performance requirements of this section.

**21 10 00 Part 2.1.2.A** The system shall be configured in a fully redundant N+1 configuration.

**21 10 00 Part 2.2.1.A** Performance shall meet the requirements of the applicable standards cited in Part 1.2.1.A.

### 2.2 Performance and Design Criteria

**21 10 00 Part 2.2.2.A** All equipment shall be rated for continuous operation at 40°C ambient temperature.

**21 10 00 Part 2.2.3.A** System efficiency at full rated load shall be a minimum of 95.0% in normal operating mode.

**21 10 00 Part 2.3.1.A** Equipment enclosure shall be constructed of minimum 1.6 mm steel sheet with IP20 minimum ingress protection per IEC 60529.

### 2.3 Physical Construction

**21 10 00 Part 2.3.2.A** Enclosure paint finish shall be RAL 7035 (Light Grey) polyester powder coating, minimum 60 micron thickness.

**21 10 00 Part 2.3.3.A** Equipment shall be suitable for installation on raised access floor with standard pedestal height of 600 mm.

**21 10 00 Part 2.4.1.A** All internal power connections shall utilise bolted joints with calibrated torque values marked on the assembly.

## PART 3 - EXECUTION

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### 3.1 Examination and Preparation

**21 10 00 Part 3.1.1.A** Prior to delivery, the Contractor shall verify that the installation area is complete and ready to receive equipment.

### 3.2 Installation

**21 10 00 Part 3.2.1.A** Installation shall be per manufacturer's written instructions and approved shop drawings.

### 3.3 Field Quality Control

**21 10 00 Part 3.3.1.A** Perform insulation resistance testing on all power cables prior to energisation. Minimum acceptable insulation resistance is 100 MΩ at 1000 VDC.

### 3.4 Commissioning and Acceptance Testing

**21 10 00 Part 3.4.1.A** The commissioning agent shall perform a complete functional test of each system.

**21 10 00 Part 3.4.2.A** Verify performance meets or exceeds the specified values in Part 2 of this section.

**21 10 00 Part 3.4.2.B** Simulate system failure and verify redundant takeover. Remaining equipment shall assume full load without interruption.

**21 10 00 Part 3.4.2.C** Perform BMS communication integration test. Verify all alarm and status signals are correctly displayed on the central BMS operator console.

**21 10 00 Part 3.4.2.D** Execute a coordinated full-load transfer test from utility to standby and back. Verify uninterrupted operation throughout the sequence.